CryptX 2.0 – University Category

Field Announcement Document

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1. Document Purpose

This document is issued to formally announce the available industry domains for CryptX 2.0 and to guide undergraduate teams in selecting, analyzing, and developing innovative solutions within a chosen domain.

2. Introduction

CryptX 2.0 is designed to challenge undergraduate students to develop innovative, scalable, and impactful solutions for real-world problems.

To support this objective, a set of industry domains has been introduced. These domains allow teams to align their ideas with current technological advancements, industry needs, and societal challenges at both local and global levels.

3. Industry Domains (Industry Selection by Participants)

Participants are required to select one industry domain from the following:

1. **Agriculture (AgriTech):** Build solutions for smart farming, crop disease detection, irrigation, yield prediction, and direct farmer-to-buyer marketplaces.
2. **Tourism, Culture & Inclusive Experiences :** Create apps for personalized travel planning, AR/VR tours, local experience discovery, and sustainable tourism tools.
3. **E-Commerce & Marketplace Solutions :** Platforms for personalized shopping, virtual try-on, smart recommendations, buyer-seller matching, trust verification, dynamic pricing, niche marketplaces, and seamless inventory/checkout experiences.
4. **Cybersecurity:** Build threat detection systems, phishing tools, vulnerability scanners, secure authentication, and data protection solutions.
5. **Disaster Management & Community Resilience :** Create early warning systems, real-time alerts, damage assessment tools, offline communication, and rescue coordination apps.

6. **Education Technology (EdTech) & Generative AI :** Develop personalized learning apps, AI tutors, gamified education platforms, and tools for teachers and students.
7. **Entertainment:** Build AI content generators, interactive games, AR/VR experiences, mood-based recommenders, and social entertainment platforms.
8. **Responsible AI, Ethics & Future of Work** : Bias detection and fairness auditing, explainable AI, privacy-preserving technologies, human-AI collaboration, reskilling tools, accessibility solutions, and inclusive digital workspaces.

These domains represent key sectors with significant relevance to Sri Lanka and global trends.

4. Industry Selection Rules

- Each team must choose only one domain
- The selected domain must remain unchanged throughout the competition
- All proposed solutions must strictly align with the selected domain
- Problem statements must clearly relate to the chosen industry

5. Objective of Industry Selection

The purpose of industry selection is to:

- Encourage domain-specific innovation
- Allow teams to work within their areas of interest or expertise
- Promote diverse solutions across multiple sectors
- Ensure solutions are relevant to real-world applications

6. Post-Selection Guidelines

After selecting an industry domain, teams are expected to:

- Identify a specific and meaningful problem within the domain
- Conduct research and analyze existing solutions

- Define a clear and concise problem statement
- Design an appropriate and feasible solution
- Plan the system architecture and technology stack

7. Innovation and Solution Requirements

Teams must ensure that their proposed solution meets the following requirements:

- The solution must not be a commonly available or widely implemented system within the selected industry
- The solution must address a real gap, limitation, or inefficiency in existing systems
- The solution must be relevant to current real-world challenges in the chosen domain
- The solution may be:
    - A completely new system, or
    - An improved version of an existing system with a clearly defined unique feature

A basic system combined with a meaningful and innovative feature is acceptable, provided it demonstrates clear value and originality.

Reusing previous project ideas is allowed only if the solution is significantly improved and aligned with the selected domain.

8. Technology and Development Flexibility

- AI may be used as an enhancement or supporting component if relevant
- Teams are free to use external APIs where appropriate
- Use of open-source code is allowed; however, teams must ensure proper understanding, modification, and originality in their implementation

9. Development Policy

- Coding and implementation are allowed
- Teams may begin development immediately after selecting their domain
- All development progress must be properly tracked and maintained

10. GitHub Repository Requirement

All teams must maintain their project code on GitHub to ensure transparency, proper version control, and easy evaluation by the organizing committee.

Requirements:

- Teams must create **one public repository** per team under any team member's personal GitHub account.
- The repository **must be set to Public** throughout the competition.
- Teams must **commit code regularly** and maintain a clear, meaningful commit history.
- Follow proper version control practices (clear commit messages, logical branching if used, etc.).
- Maintain a well-structured **README.md** file that includes:
    - Problem statement
    - Proposed solution
    - Team details (names, university, selected domain)
    - Technology stack and architecture overview (recommended)
    - Instructions on how to run the project

11. Technical Approach

Teams may adopt suitable technical approaches such as the following:

- Secure and privacy-focused systems
- Artificial Intelligence and data analytics
- Cloud-based and scalable architectures
- Digital platforms and service-based systems
- Automation and intelligent workflows

Combining multiple technologies is encouraged where appropriate.

12. Final Hackathon Day

During the final hackathon event:

- Teams will complete and finalize their solution
- Internet access will be allowed
- Teams are expected to demonstrate a working system

Evaluation will include:

- System demonstration
- Presentation or pitch

- Viva (question and answer session with judges)

13. Final Presentation & Evaluation

- Each team will deliver a presentation on the official event day:

Presentation Format:

- **15 minutes total**
    - 10 minutes for Presentation + Demo
    - 5 minutes for Technical Viva
    - A live countdown timer will be displayed during the presentation.
- Teams must strictly conclude the demo within the allocated 10 minutes.
- It is **not compulsory** for all team members to take part in the presentation.
- **Deployment is optional** and will not earn additional marks, **teams deploying must use their own valid cloud architecture plan for presentation (free-tier or trial plans are acceptable**).
- Your **cloud architecture** plan must be included as part of your **presentation**.
- **Failure to present = disqualification.**
- Judges' decisions are **final and non-negotiable**.
- Any attempt to influence judges will result in disqualification.

14. Evaluation Criteria

The projects will be evaluated based on the following criteria:

- **60% Technical Core** — Problem Identification, Innovation, Technical Implementation & System Design
- **30% Advanced Concepts** — Cloud Usage, Code Quality & Security, Entrepreneurial Value
- **10% Communication** — Presentation Quality & Clarity, Technical Viva (Q&A)

Criteria	Description	Marks
Problem Identification & Relevance	How well the team identifies a real-world, meaningful, and industry-relevant problem	20
Innovation & Uniqueness	Creativity and originality of the proposed solution	20
Technical Implementation & System Design	Quality of architecture, design decisions, and overall technical approach	10
Cloud Usage	Effective use of cloud services (e.g., AWS) with a clear plan	10
Code Quality & Security	Clean, maintainable code and proper use of secure coding practices	10
Entrepreneurial Value	Business viability, monetization strategy, and market potential	10
Presentation Quality & Clarity	Clear explanation of the idea, system flow, and effective demo delivery	10
Technical Viva (Q&A)	Ability to confidently answer technical and design-related questions	10
Total		100

15. Rules and Compliance

- Teams must strictly follow their selected industry domain.
- All work must be original.
- Solutions must be original and created by the team. Plagiarism or pre-built solutions will lead to disqualification.
- Teams must adhere to ethical and legal standards.
- Organizers reserve the right to verify submissions.
- All decisions made by the judging panel are final and cannot be challenged